DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 18

🚀 Chapter 18: GNSS

Global Navigation Satellite System — GPS, GLONASS, Galileo & Augmentation

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2. Satellite Orbits & WGS84
3. GPS — Three Segments
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1. Introduction & Systems

GNSS provides global navigation with precision measured in metres. Two fully operational systems exist; more are under development. GNSS is intended to eventually replace all terrestrial radio navigation aids.

Chapter

18

Global Navigation Satellite System (GNSS)

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Fig 18.1: GNSS overview — GPS, GLONASS, Galileo, Beidou constellations

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📡 Current & Planned GNSS Systems

System	Operator	Status
NAVSTAR/GPS	USA (DoD)	Fully operational; ICAO standard
GLONASS	Russia	Fully operational
Galileo	European Union	Under development (planned full ops ~2020+)
Beidou/Compass	China	Under development (planned ~2020)

⚡ **Why Galileo? European Sovereignty** Europe developed Galileo because access to full GPS/GLONASS capability is outside European control. A European system provides independent navigation capability not subject to US or Russian military decisions.

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2. Satellite Orbits & WGS84

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Fig 18.2: GPS satellite constellation — 24 SVs in 6 orbital planes, each inclined 55° to equator

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📡 Kepler's Laws Applied to GNSS

- Satellites (Space Vehicles, SVs) calculate their own position using Keplerian laws
- Each SV's computed orbital position = **ephemeris**
- Orbits affected by: sun/moon/planetary gravity + solar radiation → ephemeris errors

📡 WGS84 — World Geodetic Survey 1984

- GPS uses **WGS84** as its earth model (ellipsoid) — ICAO standard for aeronautical positions
- GLONASS uses PZ90; Galileo uses ETRS89 — all mathematically interconvertible
- WGS84 maximum vertical discrepancy from mean sea level: **~50 m**
- Therefore: GPS altitude **cannot be used in isolation** for vertical navigation (except medium/high cruise)

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3. GPS — Three Segments

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Introduction

The development of space based navigation systems commenced in the 1950s with the establishment of the USA Transit system. The current generation began development in the 1970's and the next generation is already under development. It is intended that GNSS will eventually replace all terrestrial radio navigation facilities. However, despite USA assertions that this is imminent, it is unlikely to be achieved in the foreseeable future.

The current systems have brought a new dimension of accuracy to navigation systems with precision measured in metres, and where special differential techniques are used the potential is for accuracies substantially less than one metre.

At present there are two operational global navigation satellite systems (GNSS), enhancements of the existing systems under development and a planned European system. These systems are:

The NAVSTAR Global Positioning System (GPS) operated by the USA.

The Global Orbiting Navigation Satellite System (GLONASS) operated by Russia. After serious problems following the disintegration of the USSR in 1989/1990 the system is now fully operational.

Local area differential GNSS (LADGNSS) to provide improved accuracy and integrity to aircraft making airfield approaches.

Wide area differential GNSS (WADGNSS) of which the European Geostationary Navigation Overlay System (EGNOS) is the European contribution to a global augmentation system providing integrity and improved accuracy.

The European **Galileo**, which is under development and intended to provide a limited service from 2014/2015 and be fully operational by 2020. The principal reason the Europeans are developing their own system is one of internal security, since access to the full GPS or GLONASS facilities is outside European control. China is also developing its own system known as Compass or Beidou 2. The system is expected to be fully operational by 2020.

This chapter will study GPS, LADGNSS and EGNOS in detail, but it should be borne in mind that GLONASS and Galileo operate on similar principles to GPS, although there are differences in implementation.

Satellite Orbits

Johannes Kepler's laws quantified the mathematics of planetry orbits which apply equally to the orbits of satellites:

Using these laws, and given a starting point, the satellites - space vehicles (SVs) calculate their positions at all points in their orbits. The SVs' orbital position is known as **ephemeris.**

Fig 18.3: Three GPS segments — Space, Control and User

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Space Segment

Parameter	GPS (NAVSTAR)
Number of SVs	**24** (21 operational + 3 active spares)
Orbital planes	**6** planes, 4 SVs each
Orbital height	**20,180 km** (10,898 NM)
Orbital inclination	**55°** to equator
Orbital period	**11 h 56 min**
Frequencies	L1: **1575.42 MHz** (civil C/A code); L2: **1227.6 MHz** (military P code)
Clock	Atomic clocks (rubidium + caesium)

Control Segment

- **Master Control Station**: Colorado Springs, USA
- **Back-up Control Station**: Onizuka AFB, California
- **5 Monitoring Stations**: Colorado Springs, Hawaii, Ascension Island, Diego Garcia, Kwajalein
- Monitor stations check SV position and clock every **12 hours** minimum
- Send corrections to SVs: position (ephemeris) and clock errors
- Clock errors: cannot adjust atomic clocks → error is included in broadcast data

User Segment

All GPS receivers using the space segment. Three types:

- **Sequential**: 1–2 channels; scans SVs sequentially; less accurate/slower
- **Multiplex**: several channels; rapidly cycles through multiple SVs
- **All-in-view (parallel)**: dedicated channel per SV; tracks all visible SVs simultaneously; selects best 4 for fix; most accurate

4. Principle of Operation

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Position Reference System

GNSS use an earth referenced three dimensional Cartesian coordinate system with its origin at the centre of the earth.

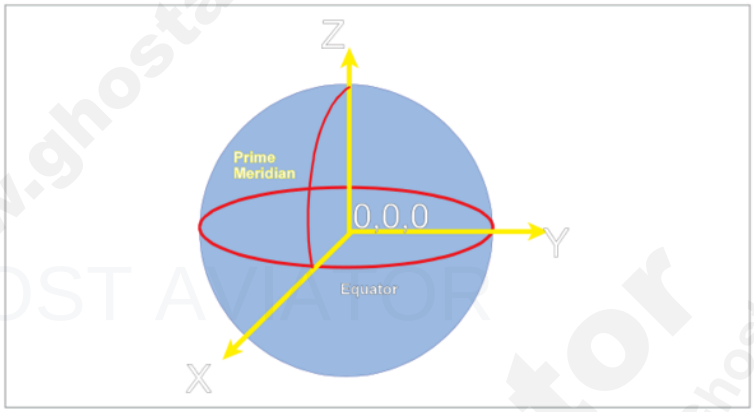


Figure 18.1

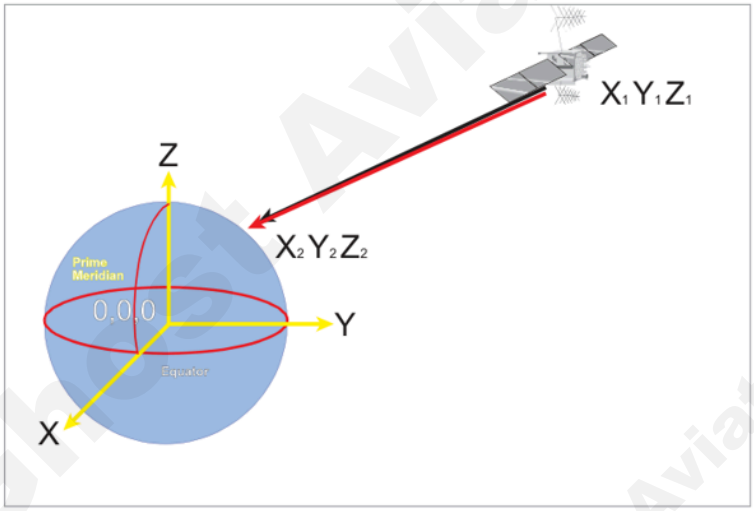


Figure 18.2

Because the systems are global, a common model of the earth was required. The World Geodetic Survey of 1984 (WGS84) was selected as the appropriate model for GPS and all GPS terrestrial positions are defined on this model and referenced to the Cartesian coordinate system. Where other models are required, for instance for the UK's Ordnance Survey maps, a mathematical transformation is available between the models (note this is incorporated as a feature of GPS receivers available in the UK). Galileo uses the European Terrestrial Reference System 1989 (ETRS89) and the Russian model for GLONASS is known as Parameters of the Earth 1990 (PZ90). WGS84 is the ICAO standard for aeronautical positions, however, since all these systems are mathematical models, transposition from ETRS89 to WGS84, for example, is a relatively simple mathematical process. Mathematically all these models are regular shapes, known as ellipsoids.

Fig 18.13: Pseudo-random code time measurement — receiver generates identical code; time offset = range

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GPS Navigation Message

- Each frame: 5 sub-frames × 6 seconds = **30 seconds** total
- Sub-frame 1: SV clock error correction
- Sub-frames 2&3: SV ephemeris data
- Sub-frame 4: ionospheric model, GPS time/UTC correlation
- Sub-frame 5: constellation almanac data (25 frames = **12.5 minutes** for full almanac download)
- Almanac: valid 4 hours to several months; usually downloaded hourly

Position Determination — Pseudo-Range

1. SV broadcasts pseudo-random noise (PRN) code on L1 (C/A) or L2 (P)
2. Receiver generates identical code and compares phase shift → measures time delay
3. Time delay × speed of light = **pseudo-range** (not yet corrected for receiver clock error)
4. Each pseudo-range = position on surface of sphere with radius >10,900 NM
5. **2 SVs**: circle (intersection of 2 spheres)
6. **3 SVs**: 2 position possibilities (one in space, one on earth) → 3D fix possible but less accurate
7. **4 SVs**: 4 simultaneous equations, 4 unknowns (X, Y, Z, T) → **true 3D fix + accurate time**

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The ellipsoids cannot be a perfect representation, nor can they represent geographical features, e.g. mountains and land depressions. The distance of mean sea level from the centre of the earth depends on gravitational forces which vary both locally and globally. Hence mean sea level will not necessarily coincide with the surface of the ellipsoid. The maximum variation between mean sea level and the surface of the ellipsoid for WGS84 is approximately 50 m. Hence the vertical information provided by any system referenced to this model cannot be used in isolation for vertical positioning, except when in medium/high level cruise with **all** aircraft using the GNSS reference and in LADGNSS applications - (where the vertical error is removed).

The GPS Segments

GPS comprises three segments:

- The Space Segment
- The Control Segment and
- The User Segment

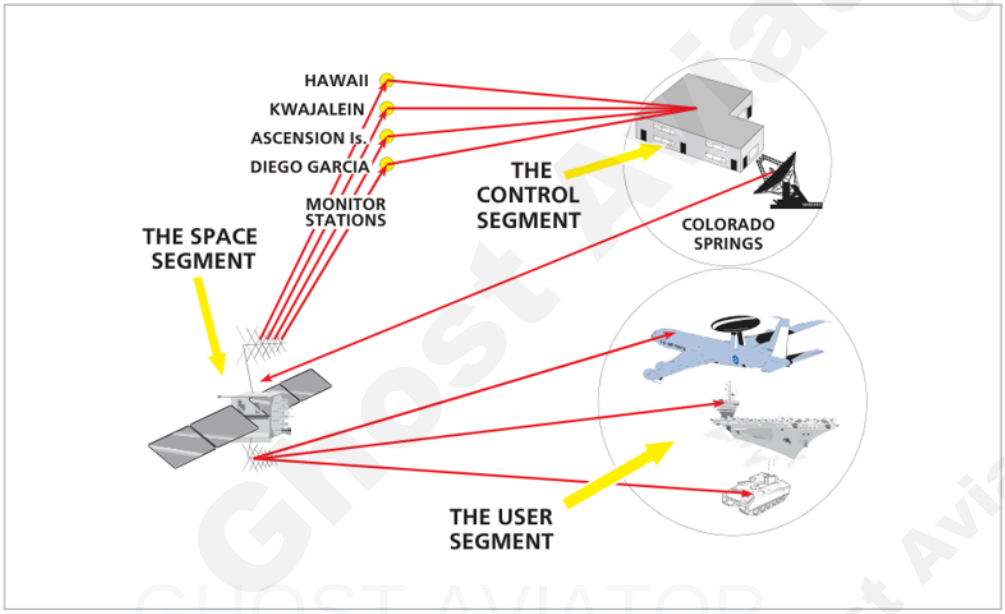


Figure 18.3 The three segments of the GPS operational control

GPS time is measured in weeks and seconds from 00:00:00 on 06 January 1980 UTC. An epoch is 1024 weeks after which the time restarts at zero. GPS time is referenced to UTC but does not run in direct synchronization, so time correlation information is included in the SV broadcast. In July 2000 the difference was about 13 seconds.

Fig 18.14–18.17: Progressive fix determination — 2 SVs circle, 3 SVs two points, 4 SVs unique 3D+time fix

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⚡ **Why 4 SVs? Receiver Clock Error** Receiver clock (crystal oscillator) is less accurate than SV atomic clocks → pseudo-ranges contain receiver clock error. A 4th SV provides a 4th equation to solve for the 4th unknown (time error T) → eliminates receiver clock error from the fix. Result: accurate 3D position AND an accurate time reference (4D fix).

⚡ **3D Fix with Altitude Input** Some receivers can produce a 3D fix using only 3 SVs + barometric altitude input. The altitude simulates a 4th SV at the earth's centre. Less accurate than a true 4-SV fix.

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5. GPS Errors

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The Space Segment

The operational constellation for GPS is specified as comprising 24 SVs. (Currently the USA has 31 SVs providing a navigational service). The orbits have an average height of 10 898 NM (20 180 km) and have an orbital period of 12 hours. The orbital planes have an inclination of 55° and are equally spaced around the equator. The spacing of the SVs in their orbits is such that an observer on or close to the surface of the earth will have between five and eight SVs in view, at least 5° above the horizon. The SVs have 3 or 4 atomic clocks of caesium or rubidium standard with an accuracy of 1 nanosecond.

An SV will be **masked** (that is not selected for navigation use) if its elevation is less than 5° above the horizon.



Figure 18.4 The GPS Satellite Constellation

The SVs broadcast **pseudo-random noise (PRN)** codes of one millisecond duration on two frequencies in the UHF band and a NAV and SYSTEM data message. Each SV has its own unique code.

L1 Frequency: 1575.42 MHz transmits the coarse acquisition (C/A) code repeated every millisecond with a modulation of 1.023 MHz, the precision (P) code, modulation 10.23 MHz repeats every seven days and the navigation and system data message at 50 Hz. The navigation and system data message is used by both the P and C/A codes.

L2 Frequency: 1227.6 MHz transmitting the P code. The second frequency is used to determine ionospheric delays.

L3 Frequency: 1381.05 MHz has been allocated as a second frequency for non-authorized users and its use is the same as the L2 frequency.

Fig 18.18: PDOP — good geometry (SVs spread out) vs poor geometry (SVs clustered) — good geometry gives smaller fix uncertainty

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⚠ GPS Error Sources (All at 95% level)

Error Source	Max Error	Detail
Ephemeris error	2.5 m	SV position error from gravitational & solar radiation effects; corrected every 12 h
SV clock error	1.5 m	Atomic clock drift; corrected every 12 h; included in broadcast
Ionospheric delay	Most significant	Ionospheric particles slow radio energy → over-reads range. Model in receiver corrects most. Worse: higher ionization, lower elevation SVs, single frequency. Dual-frequency (L1+L2) eliminates ~99% of error.
Tropospheric delay	Small	Refraction in lower atmosphere; modelled; worse at low elevation angles
Multipath	Variable	Signal reflected from buildings/terrain arrives late → longer apparent range
PDOP (geometry)	Multiplicative	Poor SV geometry multiplies all other errors. Best: 1 SV overhead + 3 near horizon 120° apart
Receiver clock	Eliminated by 4th SV	Crystal oscillator; corrected by solving 4 simultaneous equations
Selective Availability (SA)	Now 0	Deliberate USAF degradation of civil signal; cancelled in 2000; no longer an error source
Aircraft manoeuvre	Temporary	Wing/fuselage may shadow an SV → temporary accuracy degradation until another SV selected

DOP Types

DOP	Meaning
HDOP	Horizontal Dilution of Precision (X, Y errors)
VDOP	Vertical Dilution of Precision (Z errors)
PDOP	Position DOP = combination of HDOP + VDOP
TDOP	Time DOP (timing errors)

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GDOP	Geometric DOP = PDOP + TDOP combined

6. System Accuracy & RAIM

✓ ICAO GPS Accuracy Specification (SPS — Standard Positioning Service)

Parameter	Accuracy (95%)
Horizontal	**±13 m**
Vertical	**±22 m**
Time	**40 nanoseconds**

RAIM — Receiver Autonomous Integrity Monitoring

- With only 4 SVs providing 3D fix, there is NO way to detect if one SV is providing bad data
- RAIM uses additional SVs to cross-check fix consistency
- Minimum SVs for fault **detection**: **5**
- Minimum SVs for fault **exclusion** (remove bad SV): **6**
- ICAO integrity requirement: **2 seconds** warning for precision approach; **8 seconds** for non-precision
- If insufficient SVs for RAIM: GPS should not be used for primary navigation (flag generated)

7. Differential GPS (DGPS)

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	NAVSTAR – USA	GLONASS – USSR	Galileo – EU
No. of SVs:	24 SVs	24 SVs	30 SVs
Orbits:	6 Orbits	3 Orbits	3 Orbits
Orbit Height:	20 180 km	19 099 km	23 222 km
	(10 898 NM)	(10 313 NM)	(12 539 NM)
Orbit Inclination:	55° to equator	65° to equator	56° to equator
Orbit Time:	11 h 56 m	11 h 15 m	14 h 8 m
Frequencies:	L1: 1575 MHz	L1: 1600 MHz	E1: 1559 - 1591 MHz
	L2: 1227 MHz	L2: 1250 MHz	E5: 1164 - 1215 MHz
			E6: 1260 - 1300 MHz
Codes:	L1: P & C/A	L1: P & C/A	
	L2: P	L2: P	
Geoid:	WGS 84	PZ 90	ETRS 89

Figure 18.6 GNSS Systems Comparison

The Control Segment

The GPS control segment comprises:

- A Master Control Station
- A Back-up Control Station
- 5 Monitoring Stations

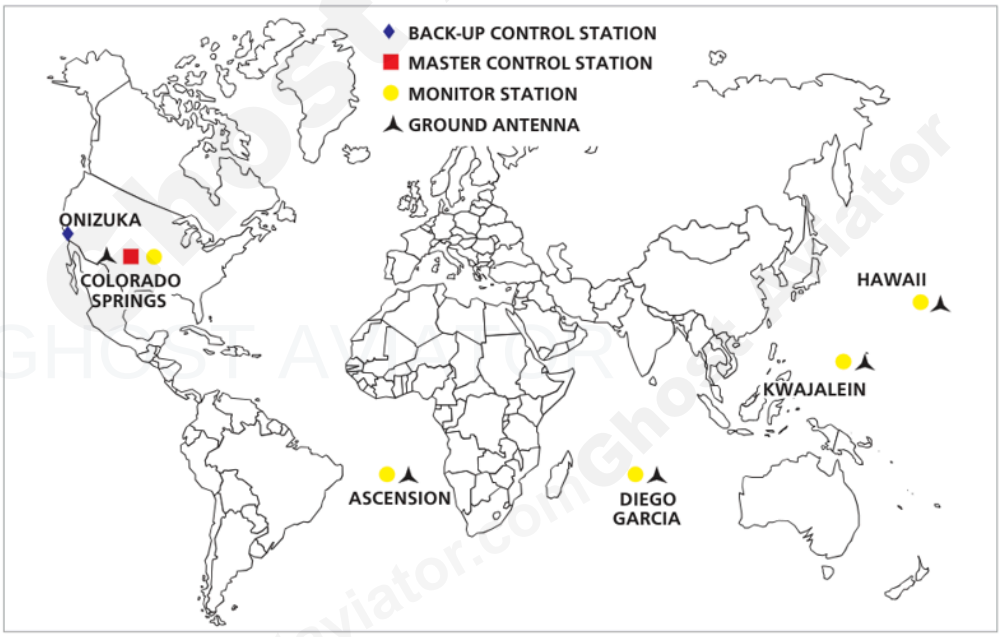


Figure 18.7 GPS Operational Control Segment

Fig 18.19: LAAS (GBAS) — precisely surveyed ground reference station detects SV errors; transmits corrections to aircraft via VHF

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📡 DGPS Concept

A precisely surveyed ground reference station measures its GPS position continuously. Since its true position is known, any GPS error = computed position − known position. Corrections are transmitted to aircraft to improve their GPS position accuracy.

- Eliminates common errors: ionospheric/tropospheric delay (virtually identical paths for ground + aircraft when close), ephemeris, SV clock
- Two implementations: LAAS (local area) and SBAS (wide area)

📡 LAAS — Local Area Augmentation System (GBAS)

- Precisely surveyed site **on the aerodrome**
- GPS receiver computes position → compares with known position → generates corrections
- Corrections transmitted to aircraft via **dedicated VHF data link**
- A **pseudolite** (pseudo-satellite) provides range to runway threshold
- When aircraft close to LAAS: ionospheric & tropospheric paths virtually identical → errors effectively eliminated
- Potential: **CAT IIIC** operations

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8. SBAS — Satellite Based Augmentation System

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The monitoring stations check the SVs' internally computed position and clock time at least once every 12 hours. Although the calculation of position using Keplerian laws is precise, the SV orbits are affected by the gravitational influences of the sun, moon and planets and are also affected by solar radiation, so errors between the computed position and the actual position occur. When a positional error is detected by the ground station, it is sent to the SV for the SV to update its knowledge of position. Similarly if an error is detected in the SV clock time this is notified to the SV, but since the clocks cannot be adjusted, this error is included in the SV broadcast.

The User Segment

The User Segment is all the GPS receivers using the space segment to determine position on and close to the surface of the earth. These receivers may be stand-alone or be part of integrated systems.

There are several types of receiver:

Sequential receivers which use one or two channels and scan the SVs sequentially to determine the pseudo-ranges.

Multiplex receivers may be single or twin channel and are able to move quickly between SVs to determine the pseudo-ranges and hence have a faster time to first fix than the sequential receivers.

Multi-channel receivers monitor several SVs simultaneously to give instant positional information. These include 'all-in-view' receivers which monitor all the SVs in view and select the best 4 to determine position. Because of the speed of operation these are the preferred type for aviation.



Figure 18.8 GPS Receiver, Control Unit

Fig 18.20: EGNOS segments — reference stations, regional control stations, master control, geostationary SVs

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SBAS (Wide Area DGPS)

- Wide-area network of precisely surveyed reference stations across a region
- Each reference station determines GPS errors for its area
- Corrections sent to Master Control Station → uplinked to geostationary satellites
- Geostationary SVs broadcast corrections to all users in region
- Geostationary SV: orbit period **24 hours**; altitude **35,800 km**; equatorial orbit

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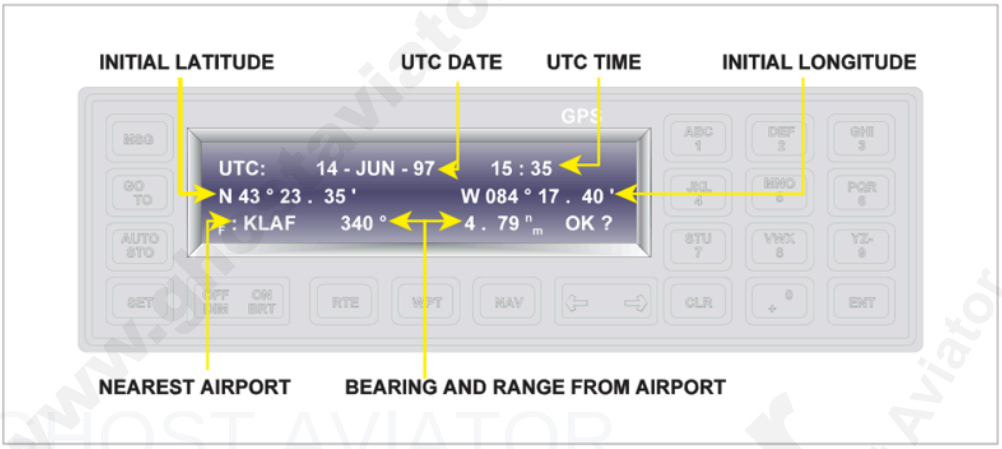


Figure 18.9 Initialization Page

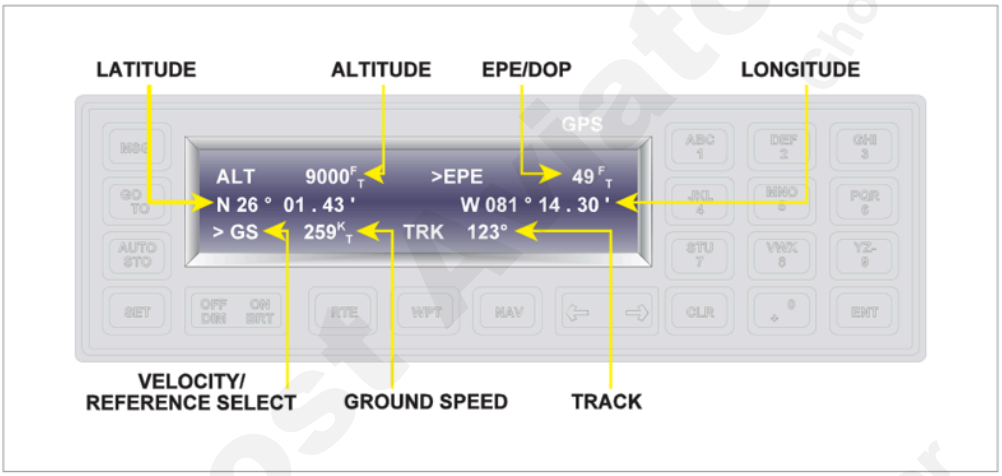


Figure 18.10 Position Page

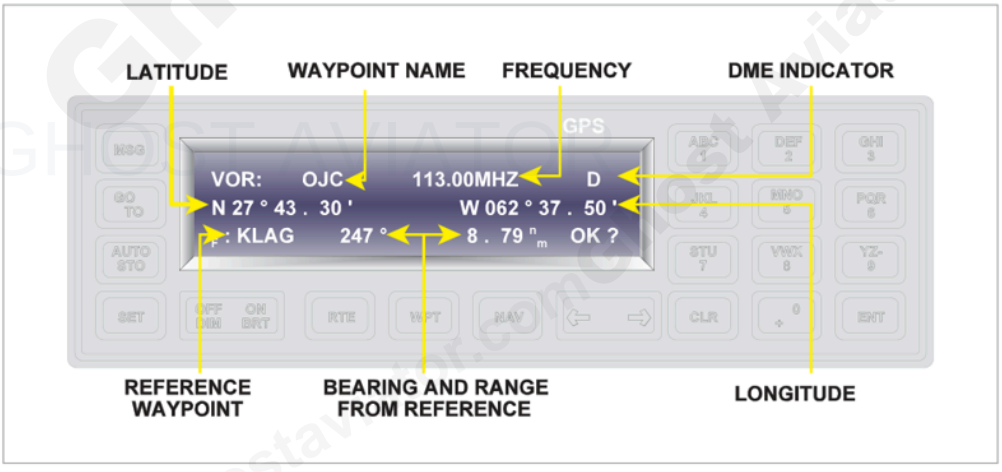


Figure 18.11 Waypoint Definitions Page

Fig: SBAS augmentation — geostationary SV broadcasts corrections; also acts as additional ranging source

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SBAS Implementations

System	Region	Name
WAAS	USA	Wide Area Augmentation System
EGNOS	Europe	European Geostationary Navigation Overlay Service
MSAS	Japan	MTSAT Satellite Augmentation System
GAGAN	India	GPS-Aided Geo Augmented Navigation

9. GNSS Systems Comparison

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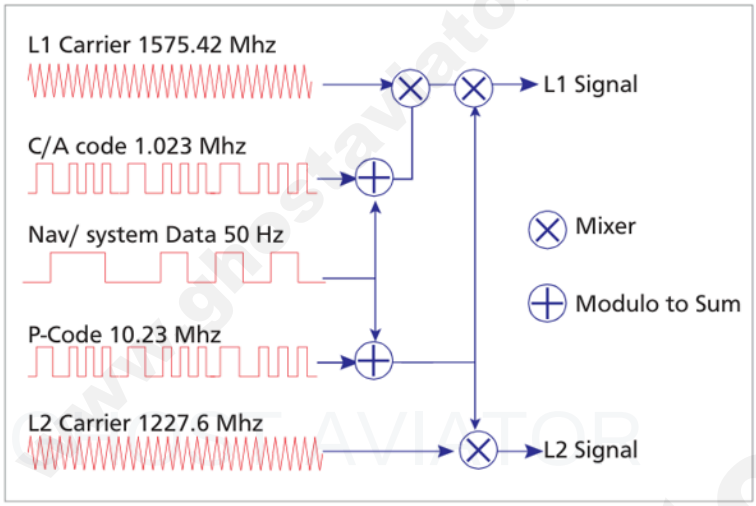


Figure 18.5

Only the C/A code is available to civilian users. The reason the use of two frequencies is important will be discussed in GNSS errors. The P code is provided for the US military and approved civilian users and foreign military users at the discretion of the US DOD. The P code is designated as the Y code when anti-spoofing measures are implemented. The Y code is encrypted and therefore only available to users with the necessary decryption algorithms.

The PRN codes provide SV identification and a timing function for the receiver to measure SV range.

The information contained in the nav and system data message is:

- SV position
- SV clock time
- SV clock error
- Information on ionospheric conditions
- Supplementary information, including the almanac (orbital parameters for the SVs), SV health (P-code only), correlation of GPS time with UTC and other command and control functions.

The two services provided are:

- The standard positioning service (SPS) using the C/A code
- The precise positioning service (PPS) using the C/A and P codes

GLONASS also has an operational constellation of 24 SVs positioned in three orbital planes inclined at 65° to the equator. The orbital height is 10313 NM (19099 km) giving an orbital period of 11 hours 15 minutes. As in GPS, GLONASS transmits C/A and P codes. The codes are the same for all SVs, but each SV uses different frequencies. The L1 frequency is incremental from 1602 MHz and the L2 frequency from 1246 MHz.

Fig 18.6: GPS vs GLONASS vs Galileo — orbits, heights, inclinations, frequencies, earth models compared

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📡 GPS vs GLONASS vs Galileo

Parameter	GPS (NAVSTAR)	GLONASS	Galileo
Operator	USA	Russia	EU
No. of SVs	**24**	24	30
Orbital planes	6	3	3
Orbital height	**20,180 km** (10,898 NM)	19,099 km	23,222 km
Inclination	**55°**	65°	56°
Orbit period	**11 h 56 min**	11 h 15 min	14 h 8 min
Civil frequency	L1: **1575.42 MHz**	L1: 1600 MHz	E1: 1559–1591 MHz
Military freq	L2: 1227.6 MHz	L2: 1250 MHz	E6: 1260–1300 MHz
Earth model	**WGS84**	PZ90	ETRS89

⚡ **Combined GPS + GLONASS** Combining both systems: improved accuracy + enhanced integrity monitoring (more SVs available). However: different earth models (WGS84 vs PZ90) must be converted — software handles this automatically.

10. GNSS Summary

Parameter	Detail
GPS constellation	24 SVs, 6 planes, 20,180 km, 55°, 11h 56m
GPS frequencies	L1 1575.42 MHz (civil C/A); L2 1227.6 MHz (military P)
Earth model	WGS84 (ICAO standard)
Control segment	Master + Backup control + 5 monitoring stations
Fix requirement	4 SVs for 3D fix + time (4D). 3 SVs + altitude input for 3D only
RAIM detection	5 SVs minimum
RAIM exclusion	6 SVs minimum

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Almanac download	12.5 minutes (25 frames × 30 s)
GPS accuracy (ICAO)	Horizontal ±13 m; Vertical ±22 m; Time 40 ns (all 95%)
Biggest GPS error	Ionospheric propagation delay (most significant in single-frequency)
SA	Selective Availability — cancelled May 2000; no longer active
LAAS/GBAS	Local area DGPS; VHF data link; pseudolite; potential CAT IIIC
SBAS	WAAS (USA), EGNOS (Europe), MSAS (Japan); corrections via geostationary SV at 35,800 km
GPS vertical	Not usable for MDA in non-precision approaches; use barometric altimetry

11. Practice Questions

Q1. NAVSTAR/GPS operates in the ___ band; receiver determines position by ___:

(a) UHF, range position lines

(b) UHF, secondary radar principles

(c) SHF, secondary radar principles

(d) SHF, range position lines

> **Answer: (a)**
> GPS L1 (1575 MHz) and L2 (1227 MHz) are in the **UHF band** (300 MHz–3 GHz). Position determined by **pseudo-range position lines** (spheres of position from each SV).

Q2. The GPS control segment comprises:

(a) space segment, user segment and ground segment

(b) ground segment and INMARSAT geostationary satellites

(c) master control station, back-up control station and five monitoring stations

(d) master + back-up control stations, five monitoring stations and INMARSAT

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Answer: (c)
GPS control segment: **Master Control Station + Back-up Control Station + 5 monitoring stations**.

Q3. Orbital height and inclination of GPS constellation:

(a) 20,180 km, 65°

(b) 20,180 km, 55°

(c) 19,099 km, 65°

(d) 19,099 km, 55°

Answer: (b)
GPS: orbital height **20,180 km**, inclination **55°** to equator.

Q4. Earth model used for NAVSTAR/GPS:

(a) WGS90

(b) PZ90

(c) WGS84

(d) PZ84

Answer: (c)
GPS uses **WGS84** (World Geodetic Survey 1984) — the ICAO standard for aeronautical positions.

Q5. Minimum satellites required for a 3D fix:

(a) 3

(b) 4

(c) 5

(d) 6

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Answer: (b)

4 SVs are required for a true 3D fix. The 4th SV provides the equation needed to eliminate receiver clock error (the 4th unknown: X, Y, Z, T).

Q6. GPS operational constellation comprises how many satellites:

(a) 12

(b) 21

(c) 24

(d) 30

Answer: (c)

GPS operational constellation: **24 SVs** (21 active + 3 spares) in 6 orbital planes, 4 per plane.

Q7. Most accurate fixing information from:

(a) four satellites clustered on one side of the sky

(b) four satellites equally spaced in azimuth at same elevation

(c) one satellite overhead, three near horizon 120° apart

(d) four satellites at high elevation angles

Answer: (d)

Best PDOP geometry: **one SV directly overhead + three SVs close to the horizon spaced 120° apart**. This maximizes the angle of cut between position spheres.

Q8. Most significant GPS error source (single frequency):

(a) PDOP

(b) receiver clock

(c) ionospheric propagation

(d) ephemeris

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Answer: (c)

Ionospheric propagation delay is the most significant GPS error for single-frequency (L1-only) receivers. It causes the signal to slow down, over-estimating range.

Q9. Frequency available to civil (non-authorized) GPS users:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

Answer: (b)

Civil GPS users have access to **L1: 1575.42 MHz** (C/A code). L2: 1227.6 MHz is military (P code). 1602 MHz is GLONASS.

Q10. PRN codes are used to:

(a) identify the satellites

(b) pass the almanac data

(c) pass navigation and system data

(d) pass ephemeris and time information

Answer: (a)

PRN (Pseudo-Random Noise) codes are used to **identify the satellites** and measure signal travel time (pseudo-range). Navigation data (almanac, ephemeris, clock corrections) is modulated on top of the PRN codes.

Q11. Minimum satellites for Receiver Autonomous Integrity Monitoring (RAIM):

(a) 3

(b) 4

(c) 5

(d) 6

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Answer: (c)
RAIM requires a minimum of **5 SVs** for fault detection. 6 SVs are needed for fault exclusion (identifying and removing the faulty SV).

Q12. Time to download the full GPS almanac:

(a) 2.5 minutes

(b) 12.5 minutes

(c) 25 minutes

(d) 15 minutes

Answer: (b)
Full almanac = 25 frames × 30 seconds per frame = **12.5 minutes**. Sub-frame 5 carries 1 page of almanac per frame; 25 frames complete the constellation almanac.

Q13. LAAS and WAAS remove errors caused by:

(a) propagation, selective availability, satellite ephemeris and clock

(b) selective availability, satellite ephemeris and clock

(c) PDOP, selective availability and propagation

(d) receiver clock, PDOP, satellite ephemeris and clock

Answer: (b)
DGPS removes: **propagation delay, ephemeris, and SV clock errors**. SA (selective availability) was cancelled in 2000. PDOP and receiver clock are not corrected by DGPS.

Q14. Most accurate satellite fixing information from:

(a) NAVSTAR/GPS & GLONASS

(b) TRANSIT & NAVSTAR/GPS

(c) COSPAS/SARSAT & GLONASS

(d) NAVSTAR/GPS alone

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Answer: (a)
Combining **GPS and GLONASS** provides more SVs in view → better geometry (PDOP) → improved accuracy and integrity monitoring.

Q15. A LAAS requires:

(a) accurately surveyed site + INMARSAT link to pass X,Y,Z corrections

(b) accurately surveyed site + INMARSAT link to pass range corrections

(c) accurately surveyed site + pseudolite to pass range corrections

(d) accurately surveyed site + pseudolite to pass X,Y,Z corrections

Answer: (c)
LAAS: **precisely surveyed aerodrome site** + corrections transmitted via **VHF data link**. A **pseudolite** provides runway threshold ranging. Corrections are pseudo-range corrections (not X,Y,Z).

Q16. GPS position errors include:

(a) selective availability, sky wave interference, PDOP

(b) propagation, selective availability, ephemeris

(c) PDOP, static interference, instrument

(d) ephemeris, PDOP, siting

Answer: (b)
GPS errors: **ionospheric/tropospheric propagation, SA (historically), ephemeris, SV clock, PDOP, multipath, receiver clock**. Sky wave, static interference, and siting are ground-based navaid errors.

Q17. EGNOS is:

(a) proposed European satellite navigation system

(b) a LAAS

(c) a WAAS

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(d) system to remove geoid-ellipsoid differences

Answer: (c)
EGNOS (European Geostationary Navigation Overlay Service) is Europe's **SBAS/WAAS** — wide area DGPS using geostationary satellites to broadcast corrections.

Q18. PRN codes are used to determine range by:

(a) measuring phase difference between received and generated code

(b) measuring Doppler shift

(c) measuring signal amplitude

(d) comparing carrier phase

Answer: (a)
The receiver generates an identical PRN code and aligns it with the received code. The **time offset (phase difference)** between the two codes = signal travel time = pseudo-range.

Q19. GPS navigation data is transmitted at:

(a) 50 Hz

(b) 1 kHz

(c) 50 bps

(d) 1575 MHz

Answer: (b)
GPS navigation data (ephemeris, almanac, clock corrections) is transmitted at **50 bits per second (50 Hz)** modulated onto the PRN codes.

Q20. Geostationary satellites used by SBAS have an orbital altitude of:

(a) 20,180 km

(b) 19,099 km

(c) 35,800 km

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(d) 23,222 km

Answer: (b)
Geostationary satellites orbit at **35,800 km** (geosynchronous orbit) in the equatorial plane with a 24-hour period. Not 20,180 km (GPS) or 19,099 km (GLONASS).

Q21. The initial range calculation is called pseudo-range because it is not corrected for:

(a) receiver clock errors

(b) receiver and satellite clock errors

(c) receiver & satellite clock errors + propagation errors

(d) receiver & satellite clock errors + ephemeris errors

Answer: (a)
Pseudo-range is initially uncorrected for **receiver clock error**. The receiver's crystal oscillator has deliberate offset error; this is solved by the 4th SV equation. SV clock error is included in the broadcast and corrected before computing pseudo-range.

Q22. Navigation and system data message transmitted through:

(a) 50 Hz modulation

(b) C/A and P PRN codes

(c) C/A code only

(d) P code only

Answer: (a)
Navigation data is transmitted at **50 bits per second** modulated on both L1 C/A and L2 P codes. The data frame structure carries ephemeris, almanac, ionospheric model, and clock corrections.

Q23. An all-in-view receiver:

(a) informs operator all required satellites are available

(b) checks all SVs in view and selects 4 with best geometry

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(c) requires 5 SVs for a 4D fix

(d) uses all visible SVs for fixing

> **Answer: (b)**
> An all-in-view receiver tracks all visible SVs simultaneously (dedicated channel per SV) and **selects the 4 with best geometry** (lowest PDOP) for the navigation fix.

Q24. When using GNSS for a non-precision approach, MDA is determined using:

(a) barometric altitude

(b) GPS altitude

(c) radio altimeter height

(d) either barometric or radio altimeter

> **Answer: (a)**
> GPS vertical accuracy is insufficient for MDA determination due to the WGS84/MSL discrepancy (up to 50 m). **Barometric altitude** must be used for MDA on non-precision approaches.

Q25. If an aircraft manoeuvre shadows a satellite being used for fixing:

(a) accuracy unaffected

(b) accuracy temporarily downgraded

(c) fix lost permanently

(d) RAIM activates automatically

> **Answer: (b)**
> Shadowing a SV temporarily removes it from the fix solution. The receiver may reselect from remaining visible SVs. Accuracy is **temporarily downgraded** until the SV is reacquired or another selected.

Q26. The maximum discrepancy between WGS84 ellipsoid and mean sea level is approximately:

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(a) 5 m

(b) 50 m

(c) 500 m

(d) 0.5 m

Answer: (c)

The maximum discrepancy between the WGS84 ellipsoid and mean sea level (geoid) is approximately **50 m**. This is why GPS altitude alone is insufficient for precision vertical navigation.

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